Final Project Proposal

CSE 429: Computer Vision and Pattern Recognition

Project Title: Benchmarking DINO for Drone-View Object Detection on VisDrone

Submitted to: Prof. Ahmed Gomaa

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Problem Statement

Aerial object detection from drone-mounted cameras is a high-impact task characterized by unique environmental constraints. The system must process raw aerial frames as input and output precise bounding boxes, class labels, and confidence scores for diverse targets (e.g., pedestrians, vehicles, bicycles).

The Challenge: Standard detection models often fail in UAV contexts because objects appear at **extremely small scales** (often $< 10 \times 10$ pixels), perspective shifts are extreme, and scene density is high. Reliable detection is critical for military surveillance, search and rescue, and traffic monitoring. This project evaluates whether the **DINO** (DETR with Improved DeNoising) architecture can bridge the gap between high-precision transformer logic and the specific demands of drone-view imagery.

Approach

We adopt the "Review and Implement" track, comparing three distinct architectures on the **VisDrone** dataset:

1. **DINO (Primary):** An anchor-free, end-to-end transformer detector utilizing Contrastive DeNoising (CDN). We will fine-tune the official model using pretrained COCO checkpoints.
2. **Vanilla DINO:** A baseline version without the denoising components to isolate the impact of the denoising mechanism (Ablation Study).
3. **YOLOv11:** A CNN-based real-time detector to benchmark against the current industry standard for edge-deployment.

Experiments and Results

Dataset

We will use the **VisDrone-DET2019** dataset, containing over 10,000 images and 10 object categories collected from drone cameras.

Implementation Details

- **Framework:** PyTorch, utilizing the official DINO and Ultralytics YOLOv11 repositories.
- **Training:** Due to compute constraints, we will perform prototype-level training for 20 epochs.
- **Hardware:** NVIDIA T4 (Kaggle/Colab).

Planned Experiments

Experiment	Purpose
DINO Fine-tuning	Primary performance results on drone imagery.
Vanilla Ablation	Quantify the impact of the denoising mechanism.
YOLOv11 Benchmark	Establish a baseline for real-time speed/efficiency.

Success Criteria

Success is defined by the functional implementation of the DINO pipeline on VisDrone, achieving a higher mAP than the vanilla baseline, and providing a comprehensive trade-off analysis between DINO's accuracy and YOLOv11's speed for real-world drone applications.